5. Bit length is measured by:
a. bits b. bits/s c. meters

6. The main difference between periodic and non-periodic digital signals is :

a. bandwidth b. frequencies c. amplitude

7. A block of 256 sequential 12-bit data words is transmitted serially in 0.016s. Calculate the time duration of 1 word :

a. 1.33 μs b. 62.5 μs c. 16 μs

8. A PCM transmission requires:
a. more bandwidth than AM
b. less bandwidth than AM
c. same bandwidth as AM

9. A bandwidth of:

a. 10 KHz for transmission of speech signals
b. 200 KHz for transmission of speech signals
c. 4 KHz for transmission of speech signals

10. UHF propagation takes place primarily through:
a. ground wave b. sky wave c. line of sight wave

11. Indicate which of the following system is digital:
a. PPM b. PCM c. PWM

12. A signal of 140 MHz has a bandwidth of s a bandwidth of ±20 MHz. What is the Nyquist sampling rate:

a. 80 MHz b. 160 MHz c. 40 MHz

FACULTY OF ENGINEERING TECHNOLOGY

Subject: Comm.& Data Transmission

Final Exam fall 22/23

Duration: 2 hours

Name: ....

Q1: Start your final exam with 5 marks, filling the appropriate word(s) from the box into the following two charts

analog, sin waves, nonperiodic, periodic, composite simple, analog, sine wave

1) Signal: .... → .... → .... → ....

2) Signal: .... → .... → .... → ....

Q2: choose the correct answers:

(Please write your answers into table only, otherwise your answers will be neglected)

Question	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*
Answer												

1. Radio waves can penetrate walls, this property can be considered as:
   a. Advantage   b. disadvantage   c. Advantage & disadvantage

2. Oscillation of sin wave is associated with its:
   a. cycle   b. amplitude   c. phase

3. Which of the following is independent of the medium:
   a. frequency   b. wavelength   c. speed

4. In a human voice:
   a. The number of frequencies is infinite, but the range is limited
   b. The number of frequencies is finite, but the range is limited
   c. The number of frequencies is infinite, but the range is unlimited

Q4: For the 11-bit binary string 01001100011:

a) Find the number of transitions for the encoding schemes Manchester & Differential Manchester

b) If the above combination rate is 10Mbps, calculate the average signal rate using Manchester code

c) Calculate the number of data elements carried by each signal element in differential Manchester code

d) Calculate maximum bandwidth (11 marks)

Q3: If the data combination is represented as: 10010011011

which will be augmented according to the generator x[illegible]$+x+1$

a) find CRC

b) if the received combination is $x^{11}+x^{10}+x^{8}+x^{7}+x^{5}+x^{4}+x^{3}+x^{2}$

what is the decoding result at the receiver for this combination (show your justification)

(10marks)